Networking 2

Activity 2 - Routing Concepts — Regional Network Outage Investigation

1. Scenario: The 08:00 Regional Services Outage

You are the junior network engineer for **Northstar Health Services**. At 08:00, users at Headquarters report that they cannot reach the application server at the Disaster Recovery site. The Network Operations Center can still reach its local monitoring host, but the incident dashboard is incomplete. Management needs service restored before the next clinical reporting cycle.

The previous engineer left only a network design image and an addressing record. No Packet Tracer file is provided. You must translate the design into a working topology, configure it from your own plan, and then investigate the outage conditions introduced by the instructor. You are not given a configuration recipe. You are expected to infer the required state from the topology image, addressing plan, and business requirements, then prove that your correction works.

Your incident objectives are:

- Restore communication from the Headquarters workstation to the Disaster Recovery application host.
- Preserve the Monitoring LAN and make it reachable from both sites.
- Use static and default routing only; do not configure a dynamic routing protocol.
- Record the evidence that distinguishes an interface fault from a routing fault.
- Explain how a packet travels through the network and why each router forwards it through its selected interface.

Important constraint: This is an investigation, not a command-following exercise. The handout gives the required outcomes, addressing facts, and command families. It does **not** provide a step-by-step IOS configuration or a command-by-command solution. Choose appropriate IOS commands, test hypotheses, and justify every change.

2. Learning Objectives

By the end of this laboratory, you should be able to:

1. **Infer** a router and host configuration from a business scenario and an addressing plan.
2. **Configure** IPv4 interfaces, descriptions, hostnames, and static routes using Cisco IOS without a supplied command recipe.
3. **Diagnose** whether a failure is caused by interface state, addressing, gateway configuration, or missing routing information.

4 **Interpret** connected, local, static, and default routes, including next-hop, exit-interface, administrative-distance, and metric fields.

5 **Trace and explain** Layer 2 de-encapsulation/re-encapsulation and Layer 3 forwarding across multiple router hops.

6 **Defend** a network correction with command evidence, packet tests, and a concise incident report.

3. Technical Scope

The investigation is limited to concepts supported by the uploaded lecture and asynchronous activity: router functions, IPv4 addressing and prefixes, default gateways, connected and local routes, static routes, default routes, route selection, next hops, exit interfaces, administrative distance, metric, routing-table inspection, basic IOS configuration, ping, trace, and troubleshooting. Dynamic routing is discussed conceptually but is not configured.

4. Equipment and Initial Conditions

Use the following in Cisco Packet Tracer:

- Three Cisco 2911 routers: **R1**, **R2**, and **R3**.
- Three Layer 2 switches: **SW1**, **SW2**, and **SW3**.
- Three PCs: **PC1**, **PC2**, and **PC3**.
- Appropriate Ethernet connections.
- A text editor or worksheet for the incident report.

Create the entire Packet Tracer topology yourself from the topology image. The instructor will not provide a starter .pkt file or a partially configured topology. Save your work as Northstar_Incident_<Surname>.pkt and make a separate baseline copy when you are ready for the individual fault-injection check.

5. Topology Image

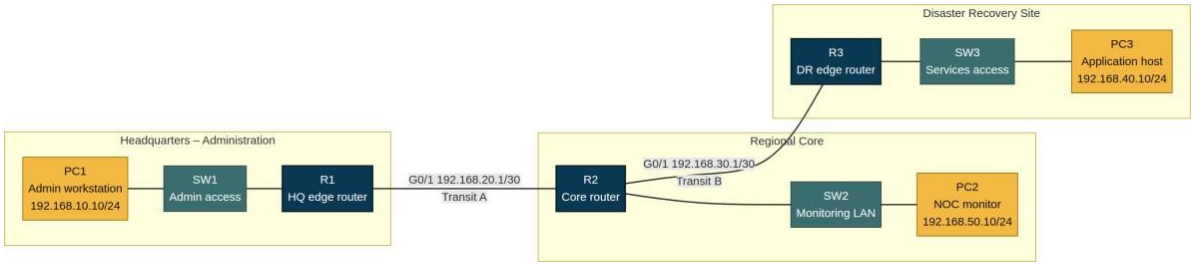


Figure 1. Northstar regional routed network. This image is the only topology reference supplied to students. Reproduce its devices, links, labels, and logical arrangement in Packet Tracer; use the tables below as the addressing authority.

5.1 Student topology-construction requirements

Build and label the topology shown in Figure 1. Do not submit a hand-drawn substitute. The completed Packet Tracer file must contain the following logical segments:

Segment	Required relationship
Headquarters LAN	PC1 connected through SW1 to R1 G0/0
Transit A	R1 G0/1 connected to R2 G0/0
Monitoring LAN	R2 G0/2 connected through SW2 to PC2
Transit B	R2 G0/1 connected to R3 G0/0
Disaster Recovery LAN	R3 G0/1 connected through SW3 to PC3

Choose the appropriate Packet Tracer cable types, switch access ports, and device placement. Record the actual switch ports and cable types in your report. The instructor may ask you to explain why each connection was selected.

5.2 Addressing authority

Device	Interface/ NIC	IPv4 address	Prefix/mask	Default gateway	Business role
R1	G0/0	192.168.10.1	/24 — 255.255.255.0	—	Headquarters gateway
R1	G0/1	192.168.20.1	/30 — 255.255.255.252	—	Transit A
R2	G0/0	192.168.20.2	/30 — 255.255.255.252	—	Transit A
R2	G0/1	192.168.30.1	/30 — 255.255.255.252	—	Transit B
R2	G0/2	192.168.50.1	/24 — 255.255.255.0	—	Monitoring gateway
R3	G0/0	192.168.30.2	/30 — 255.255.255.252	—	Transit B
R3	G0/1	192.168.40.1	/24 — 255.255.255.0	—	DR gateway
PC1	NIC	192.168.10.10	/24 — 255.255.255.0	192.168.10.1	Headquarters workstation
PC2	NIC	192.168.50.10	/24 — 255.255.255.0	192.168.50.1	NOC monitor
PC3	NIC	192.168.40.10	/24 — 255.255.255.0	192.168.40.1	DR application host

The /30 transit networks have only two usable addresses. Do not use network or broadcast addresses as host addresses.

5.3 Business reachability requirements

The finished network must satisfy these requirements:

Requirement	Required behavior
Headquarters to DR	PC1 must reach PC3.
DR to Headquarters	PC3 must reach PC1.
Monitoring visibility	PC2 must reach both PC1 and PC3.
Router knowledge	Every router must have a route for every remote LAN it must serve.
Default forwarding	R3 must have a default route toward R2 for destinations not matched by a more specific route.
Configuration scope	Use IPv4 static routing and a default static route. Do not enable RIP, OSPF, EIGRP, or another dynamic protocol.

6. Pre-Investigation Questions

Answer before making changes. Explain your reasoning rather than supplying isolated definitions.

7 PC1 sends traffic to 192.168.40.10. Which network is local to PC1, and what is PC1's first-hop device?

- PC1 belongs to 192.168.10.0/24, with IP address 192.168.10.10 and gateway 192.168.10.1. Since 192.168.40.10 belongs to a different network, PC1 sends the packet to its default gateway, R1.

8 What evidence would distinguish an administratively down interface from a missing route?

- Use show ip interface brief to inspect interface status and show ip route to inspect routing information. An administratively disabled interface typically shows administratively down. A missing route means the destination network is absent from the routing table and no suitable default route is available. Both problems can exist at the same time, so inspect both.

9 Why should R1 G0/1 and R2 G0/0 belong to the same /30 network, while R1 G0/0 must belong to a different network?

- R1 G0/1 (192.168.20.1/30) and R2 G0/0 (192.168.20.2/30) form Transit A, 192.168.20.0/30. They must be in the same subnet to communicate directly. R1 G0/0 is the Headquarters gateway and belongs to 192.168.10.0/24, a separate LAN. Separating these networks allows R1 to route traffic between them.

10 Predict the connected and local routes R2 should have after all three R2 interfaces are correctly addressed and active.

- R2 should have these connected routes:
- C 192.168.20.0/30 on G0/0.
- C 192.168.30.0/30 on G0/1.
- C 192.168.50.0/24 on G0/2.
- It should also have local routes (L) for its own interface IPs: 192.168.20.2/32, 192.168.30.1/32, and 192.168.50.1/32.

11 What is the difference between a route's next-hop address and its exit interface?

- The next hop is the IP address of the neighboring router to which a packet should be forwarded. The exit interface is the local interface through which the packet leaves.

  For example, R1 forwards traffic to the DR LAN using next hop 192.168.20.2 through G0/1.

12 Why can a more specific route be preferred over a default route?

- Routers use longest-prefix matching: among matching routes, the route with the longest prefix is preferred. For example, a route to 192.168.40.0/24 is preferred over 0.0.0.0/0 for destination 192.168.40.10. The default route is the fallback when no more specific matching route exists.

13 What changes at Layer 2 when a packet passes from R1 to R2? What does not change at Layer 3?

R1 removes the incoming Ethernet frame header and trailer, examines the IP packet, and forwards it by encapsulating the packet in a new Ethernet frame for the R2 link. The source and destination MAC addresses change for the new link. The source and destination IP addresses normally remain the same during ordinary routing, although the TTL decreases by one at each router hop.

14 Which evidence would be sufficient to claim that the routing problem is fixed rather than merely hidden by a successful local ping?

A local ping proves only that a particular local path works. Stronger evidence includes correct interface status, correct routing tables on all routers, successful PC1-to-PC3 and PC3-to-PC1 pings, successful PC2 reachability to both remote PCs, and a trace consistent with the expected path. The saved configuration and screenshots should support the claim.

7. Investigation Rules

You may use the IOS command families introduced in the lecture, including interface-status inspection, routing-table inspection, running-configuration inspection, detailed interface inspection, reachability tests, and trace tests. You may also use normal configuration-mode navigation and save commands.

You may **not** receive or copy a completed configuration block during the student attempt. Every command you use must have a stated purpose in your report. Do not change an address merely because a ping fails; first identify the layer at which the failure occurs.

Use this investigation loop:

> ***Observe → hypothesize → make the smallest justified change → verify → document.***

8. Challenge

Phase 1 — Build the student topology and establish the baseline

Create the routers, switches, PCs, links, device names, and labels from Figure 1. Save the file as Northstar_Incident_<Surname>.pkt.

Before troubleshooting any outage condition, configure and verify your own baseline network. Your baseline must satisfy every business reachability requirement. Save a second copy as Northstar_Incident_<Surname>_Baseline.pkt; this is the recovery reference used during the individual instructor check.

After building and configuring the topology, collect baseline evidence from each router and host. Determine:

- Which interfaces have the correct addresses.
- Which interfaces are administratively down or physically down.
- Which connected and local routes exist.
- Which remote networks are absent.
- Which host gateways are correct.
- Which of the required reachability tests fail.

Create a **baseline table** with columns for device, observation, evidence command or screen, interpretation, and severity.

Phase 2 — Reconstruct the intended network state

Use the addressing authority and business requirements to produce your own implementation plan. Before entering configuration mode, your plan must identify:

- The interface addresses and masks for each router.
- The interfaces that must be enabled.
- The static route destinations each router needs.
- The next hop or exit interface for each route.
- Whether R3's default route is sufficient for a particular destination or whether a more specific route is required.
- The verification test that will prove each planned change.

Submit this plan even if the supplied file already contains some correct settings.

Phase 3 — Restore service

Implement the smallest set of changes that satisfies the plan. You are responsible for selecting the correct IOS commands. Do not paste a recipe from another source.

After each logical group of changes, capture evidence showing what changed. At minimum, your evidence must include:

- Interface address and operational-state evidence from every router.
- Routing-table evidence from every router.
- Evidence of the active configuration for the interfaces and routes you changed.
- Host addressing evidence for PC1, PC2, and PC3.
- Successful local and remote reachability tests.

Phase 4 — Prove the forwarding path

Use a source and destination pair that crosses all three routers. Record the expected and observed path.

For **PC1 → PC3**, complete this table:

Hop	Incoming interface	Destination IP examined	Matching prefix	Next hop	Exit interface	Layer 2 action
R1	G0/0	192.168.40.10	192.168.40.0/24	192.168.20.2	G0/1	Removes incoming frame and creates a new frame for R2
R2	G0/0	192.168.40.10	192.168.40.0/24	192.168.30.2	G0/1	Re-encapsulates the IP packet for R3
R3	G0/0	192.168.40.10	192.168.40.0/24	Directly connected	G0/1	Creates an Ethernet frame addressed to PC3

Then explain why the Layer 2 frame changes at each hop while the destination IP remains 192.168.40.10.

Phase 5 — Individual fault-injection recovery check

This phase is completed **individually and in the presence of the instructor**. Do not modify the topology until the instructor has verified your working baseline and saved a checkpoint. The instructor will inject one fault at a time into your completed topology, observe your investigation, and restore the topology before injecting the next fault. You will not be told which fault was selected.

Possible faults are:

- A transit interface is administratively disabled.

- A transit interface has the wrong mask.
- A host has the wrong default gateway.
- A remote static route is absent.
- A static route points to the wrong next hop.
- An interface address is in the wrong subnet.

For observes your work:
each individually administered fault, complete an incident record while the instructo

Incident	Symptom	First evidence	Root cause	Minimal correction	Verification evidence	Why communication failed
1	PC1 cannot reach PC3	show ip interface brief on R1/R2	R1 G0/1 administratively down, if observed	Enter R1 G0/1 and run no shutdown	Interface becomes up/up; PC1 ping to PC3 succeeds	R1 could not forward traffic over Transit A
2	DR cannot reach Headquarters	show ip route on R3	Route to 192.168.10.0/24 missing, if observed	Add ip route 192.168.10.0 255.255.255.0 192.168.30.1	Route appears; PC3 can ping PC1	R3 lacked a usable route to the Headquarters LAN
3	PC2 cannot reach a remote network	PC2 IP Configuration and gateway ping	Incorrect PC2 gateway, if observed	Set gateway to 192.168.50.1	Gateway and remote-host pings succeed	PC2 was sending off-subnet traffic to the wrong gateway

A recovery is scored only after you identify the cause, apply the correction, verify the affected path, and explain why the evidence supports the diagnosis. The instructor must confirm that the topology is restored to the pre-injection state before the next fault begins. Do not inspect another student's fault-injection attempt.

Phase 6 — Independent extension challenge

The NOC requests a new management subnet, **192.168.60.0/24**, reachable behind R3. Add a fourth host, **PC4**, to a new LAN on R3's available interface or to an instructor-approved equivalent segment. Use address 192.168.60.10/24 and gateway 192.168.60.1.

You must decide:

15 Which R3 interface and switch connection should serve the new LAN.
   - Use R3 G0/2 connected to SW4, with PC4 connected to SW4, provided G0/2 is available.

16 Which router interfaces or routes must be added.
   - Configure R3 G0/2 as 192.168.60.1/24; configure PC4 as 192.168.60.10/24, gateway 192.168.60.1; add a route on R2 via 192.168.30.2 and on R1 via 192.168.20.2.

17 Whether each existing router needs a specific route, a default route, or both.
   - R1 and R2 need specific routes to 192.168.60.0/24. R3 gets a connected route automatically and retains its existing specific routes and default route.

18 How you will prove PC1, PC2, PC3, and PC4 can reach the required destinations.
   - Ping PC4 from PC1, PC2, and PC3; ping the other PCs from PC4; verify routes on all three routers.

Do not modify the original transit networks. Document your design assumptions and any interface-model limitation.

9. Required Analysis Questions

19 Which baseline observation most strongly indicated the original fault? Why?
   - The strongest observation is the evidence that directly identifies the fault, such as a disabled interface or missing route.

20 How did you distinguish an interface problem from a routing-table problem?
   - Check interface status and the routing table.

21 Which route on R1 is selected for 192.168.40.10, and why?
   - R1 selects 192.168.40.0/24 through 192.168.20.2.

22 Why is R2's route to 192.168.50.0/24 normally connected rather than static?
   - The Monitoring LAN is directly connected to R2.

23 What happens when a destination has no matching route and no usable default route?
   - The router drops the packet if no matching or default route exists.

24 What does [1/0] mean on a static route in the routing table?
   - [1/0] means administrative distance 1 and metric 0.

25 How would a wrong next hop differ from a wrong exit interface in the observed symptoms?
   - A wrong next hop sends traffic toward the wrong neighbor; a wrong exit interface sends it through the wrong link.

26 Why should a default route not be used as an excuse to omit a required specific route?
   - Specific routes provide explicit paths to required networks.

27 Which command or test gave the strongest proof of end-to-end success? Defend your choice.

28 What was the smallest configuration change that restored one of the injected faults? Why is minimizing changes good troubleshooting practice?

29 How does the router use the destination IP, routing table, next hop, and exit interface during forwarding?

30 What additional evidence would you collect in a production network before declaring the incident closed?

10. Submission Requirements

Submit a folder named Surname_Northstar_Routing_Incident containing:

31 The final Packet Tracer file.
32 The baseline table and implementation plan.
33 The final addressing and route tables.
34 Selected evidence from interface, configuration, and routing-table inspection.
35 The completed PC1-to-PC3 forwarding-path table.
36 Three fault-injection incident records.
37 Answers to the pre-investigation and analysis questions.
38 The independent extension design and verification results.
39 A topology screenshot with device names visible.

11. Assessment Rubric — 100 Points

Criterion	Points	Standard
Scenario analysis and baseline	15	Accurate observations, evidence, hypotheses, and prioritization.
Addressing and topology reconstruction	15	Correct interpretation of interfaces, prefixes, gateways, and links.
Routing design and implementation	20	Correct static/default routing with justified next hops and no dynamic protocol.
Verification evidence	15	Appropriate command selection, route evidence, local tests, remote tests, and trace.
Fault isolation and recovery	15	Three evidence-based diagnoses, minimal corrections, and explanations.
Packet-forwarding analysis	10	Correct route matching, next-hop/exit-interface reasoning, and encapsulation explanation.
Independent extension	5	Coherent new-LAN design and verification.
Documentation and professional practice	5	Reproducible file, organized report, clear screenshots, saved configuration, and precise writing.
Total	**100**	

12. Success Criteria

The incident is closed only when:

- All required router interfaces have the intended addresses and operational state.
- Every router has the necessary connected, local, static, and default entries.
- PC1 reaches PC3, PC3 reaches PC1, and PC2 reaches both remote hosts.
- The trace and route evidence support the claimed forwarding path.
- All three injected faults are diagnosed, corrected, and explained.
- The extension LAN is documented and tested if assigned.
- The configuration is saved and the report is reproducible by another student.